Disperse to ignite, concentrate to endure: a decay-driven crossover in modular complex contagions

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Abstract. Should a costly new practice be seeded into a few dense teams, or scattered across the organization? We study this on synthetic two-layer organizations of dense small teams, using fractional-threshold contagion (heterogeneous thresholds, innovator atom, credibility-weighted exposure) under a paired protocol: within each condition, every strategy runs on the same 50 organizations with identical agent draws. Without decay, dispersed seeding dominates the tested ignition band: +39.7 percentage points of terminal adoption over cluster seeding at the headline setting (95% CI [35.4, 43.9]), winning 14 of 40 cells of a threshold–budget map (23 saturated or starved cells equivalent within ± 2 pp, 3 uncertain, no cluster win of ≥ 2 pp). Under reinforcement-dependent decay the terminal-adoption advantage shrinks and reverses between the tested relapse rates: dispersed cascades are increasingly throttled, while clustered adoption stays near 30% across the tested decay settings. The sign of the contrast replicates on a real modular topology (email-Eu-core, synthetic behavioral attributes). All diffusion outcomes are simulated and all behavioral attributes synthetic; version-controlled code with frozen seeds reproduces every number.

Keywords: complex contagion, threshold models, seeding strategies, organizational networks, agent-based simulation

1 Introduction and related work

Threshold models [3] make costly adoption require reinforcement from multiple credible sources. Cascades on random graphs obey a vulnerability boundary [12]; complex contagions need wide bridges of overlapping ties, which long-range shortcuts destroy [2, 1]. Network-intervention taxonomies [10] stress matching the intervention to context; the expectation that clustered seeding should win follows from those wide-bridges results—evidenced largely in populations not already embedded in cohesive groups. Adjacent results strain it: modularity aids cascades only up to an optimum [6]; dispersed neighbor seeding beats clustered seeding in threshold models [7]; seed placement is its own measurement problem [4, 9]; random ties match or beat clustered ties unless interest decays fast [11, 8]—a structural reach–reinforcement trade-off whose seeding-strategy analogue we test at fixed topology, asking when decay reverses the ordering. Real organizations are modular—nearly everyone already sits inside a dense small team—so what does cluster seeding buy when every seed is already surrounded by a cluster? We answer with a transparent agent-based model under a paired protocol pre-specified in a version-controlled decision log.

2 Model and paired protocol

A generator produces two-layer organizations: a formal hierarchy (8 departments, 249 line teams of mean size 8, N=2000) and an informal influence network—dense within teams (p=0.9), locally concentrated between ring-adjacent "sister" teams, sparse across departments (silo 0.85)—on which adoption propagates. Agent i holds threshold $\theta_i \sim \mathrm{Beta}(\mu\kappa, (1-\mu)\kappa)$, μ=0.30, κ=20, and is an innovator (θ_i=0) with probability 0.025. Exposure is the credibility-weighted share of contacts whose adoption is visible,

$$\mathrm{share}_i(t) = \frac{\sum_{j \in N(i)} v_j w_{ij} a_j(t-1) + b(t)}{\sum_{j \in N(i)} w_{ij} + b(t)}, \tag{1}$$

with credibility w (close collaborator 1.0 > manager 0.7 > distant peer 0.6 > comms 0.3), source visibility v_j (default 1), and $b(t)$ a transient comms term during a broadcast. Adoption requires ready ($\mathrm{share}_i \geq \theta_i$ and $\mathrm{share}_i > 0$), willing (Bernoulli 0.85), and able (1.0 here); updates are synchronous. Decay (off unless stated): an adopter relapses with probability ρ per step while $\mathrm{share}_i < r\,\theta_i$, where r sets how much reinforcement retention requires. We report terminal adoption (adopted share among active agents at the fixed point; 98–100% converge) and, secondarily, cumulative reach (ever-adopted share). Four seeded strategies share a 5% budget (100 seeds): random, champions (top informal degree), cluster (whole teams), one-per-team (round-robin coverage); broadcast is a zero-seed awareness baseline—fixed heuristics, not influence-maximization optima [5, 7].

Paired protocol. Within each condition cell, every strategy runs on the same 50 fresh organizations with identical agent draws (common random numbers; cells may use independent samples), so contrasts are within-organization paired differences—confound-free at the organization level, with no efficiency gain over independent sampling assumed. Two confirmatory families (regime map, decay grid) were pre-specified in a version-controlled decision log—after exploratory n=12 panels, before the confirmatory run—and Holm-corrected for both difference and TOST-equivalence against a ±2 pp practical band; each cell gets a three-way verdict: win, equivalent (within ±2 pp, for the tested cells and model), or uncertain. Seeds were frozen before execution; all paired results come from a single successful full-pipeline execution (6 350 simulations).

3 Results

Without decay, dispersion dominates the ignition regime. At the headline setting, random seeding reaches 72.0 ± 9.2% terminal adoption (mean±sd) against 32.4 ± 11.0% for cluster seeding: a paired difference of +**39.7** pp, 95% CI [35.4, 43.9]. Degree targeting adds ∼3.9 pp (descriptive), and coverage with one seed per team

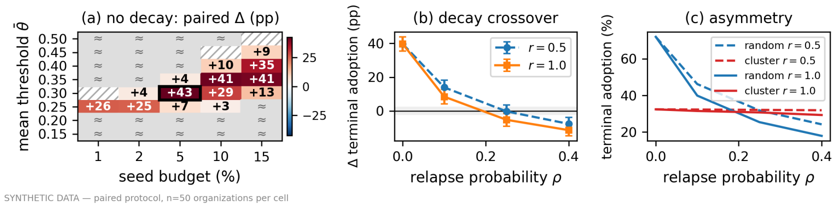

Fig. 1. Synthetic data. (a) Paired Δ terminal adoption (random − cluster, pp) over $\bar{\theta}\times$ seed budget (n=50 pairs/cell; Holm-corrected difference and TOST families): red = dispersion wins; "≈" = equivalent within ±2 pp; hatched = uncertain; box = headline setting. (b) The same contrast under decay (95% CIs; ±2 pp band). (c) Absolute levels: clustered adoption barely moves; dispersed collapses.

yields no clear difference from random (−1.4 pp, CI [−5.0, +2.2])—on average, random already lands seeds in 82 of 249 line teams (∼205 end at majority) vs. 13 (∼91) for cluster. Across the map (Fig. 1a), dispersion wins 14 of 40 cells—the diagonal ignition band, +2.8 to +42.6 pp ([39.1, 46.0] at the boxed cell, an independent paired sample)—the strategies are equivalent within ±2 pp (TOST) in 23 cells (saturation at low $\bar{\theta}$, starvation at high $\bar{\theta}$/low budget), 3 stay uncertain; no cell shows a practically relevant cluster win (≥2 pp)—the largest cluster edge, 0.8 pp in the starved corner, sits inside the band. Broadcast converts 5.5 ± 3.4% (0.0% in all 12 runs without innovators; independent panel).

Decay drives a crossover. With relapse active (Fig. 1b,c), the dispersion advantage falls monotonically in ρ and flips sign between the tested values. When retention requires full reinforcement (r=1): +8.4 pp [4.2, 12.7] at ρ=0.10, −**5.3** pp [−9.0, −1.5] at ρ=0.25, −11.4 pp [−14.8, −8.0] at ρ=0.40—the crossover lies between ρ=0.10 and 0.25. At the weaker coupling r=0.5 it comes later: +13.9 pp [9.6, 18.3] at ρ=0.10, parity still uncertain at ρ=0.25 (−0.3 pp [−4.1, +3.5]), reversal at ρ=0.40 (−7.7 pp [−11.5, −3.9]). The crossover is asymmetric: clustered adoption barely moves (32.4% at ρ=0 to 29.3% at the harshest setting)—it endures rather than wins—while dispersed adoption collapses (72.0% to 17.9%). It is also endpoint-dependent: at (r=1, ρ=0.25) the terminal contrast has reversed, but the cumulative-reach contrast (secondary, descriptive) is still uncertain (−1.8 pp [−5.4, +1.8]; clearly negative only at ρ=0.40: −7.4 pp [−10.7, −4.1])—so the terminal reversal there is driven mainly by differential retention, 0.99 (cluster) vs. 0.86 (random). Relapse throttles the dispersed cascade while it grows: random's ever-adopted share is 29.0% vs. 72% without decay.

Robustness and a real topology. The no-decay contrast keeps its sign across one-at-a-time organization variants ($N \in \{500, 2000, 8000\}$, team size $\{5, 12\}$, silo $\{0.5, 0.7, 0.95\}$; descriptive): paired Δ from +14.7 to +48.3 pp, all CIs excluding zero. On the real email-Eu-core topology (986 nodes, 42 ground-truth departments; all behavioral attributes synthetic) the contrast keeps its sign: +13.5 pp [1.0, 26.0] over 50 paired attribute draws. Adoption there is bimodal: draws end low (median 9%) or high (median 85%), none between 21% and 82%. This is a sign replication on a real modular topology, not an empirical validation of the mechanism.

4 Discussion

Once everyone already lives in a dense team, guaranteed local reinforcement—cluster seeding's defining asset—is no longer scarce; scattering buys more ignition surface per seed, and decay then decides whether dispersed ignitions compound into organization-wide growth: relapse throttles the growing cascade, while clustered adoption sits on local critical mass relapse barely touches. The prescription is conditional and asymmetric: disperse to ignite when adoption reinforces itself; concentrate to endure when reinforcement erodes quickly. A local-ignition predictor, pre-specified in the same log, failed its validation gates (AUC 0.64, below even the 0.70 partial threshold): most dispersed-seeding ignition is not explained by the local predictor; a two-level model is future work. All of this holds within a synthetic, uncalibrated model family—thresholds unobserved, time abstract, decay parameters stipulated on a grid; a 21-entry limitations file ships with the code.

Reproducibility. Every modeling decision (D1–D23) is logged with alternatives; analysis families pre-specified there; seeds frozen; one successful full-pipeline execution for the paired results. CI fails if any published number changes; every statistic is machine-extracted and audited. Repository: AGPL-3.0, public at publication.

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